

VERIQO Commercial Readiness Report

This is Commercialization Sprint Phase 9 (item 23); the single source of truth the reviewer's item 13 (Commercial Readiness Matrix) and item 25 (one-page executive summary) ask for, plus item 21's documentation inventory and item 22's Commercialization Gate walkthrough. Every status below cites the real code, test, or document backing it -- nothing here is asserted without something to point at.

1. Executive Summary -- VERIQO COMMERCIALIZATION GATE

(Item 25's exact requested format.)

Area	Status
CORE TRUST PIPELINE	GREEN
EVIDENCE FABRIC	GREEN
DECISION TRUST	GREEN
ACTION AUTHORIZATION	GREEN
DOSSIER	GREEN
REPLAY	GREEN
INDEPENDENT VERIFIER	GREEN -- and, since P0-E, performing real Ed25519 signature and key-state verification against a caller-supplied trusted-key registry, not a permanent signature SKIP (see §2).
TENANT ISOLATION	GREEN -- structural isolation inside the Store is real and adversarially tested, and since P0-B the effective tenant is derived from the verified JWT subject via <code>pkg/commercial/tenancy.Membership</code> , not from a caller-supplied field (see §2, Tenant Isolation row).
SECURITY	GREEN/YELLOW -- JWT/RBAC/API-key/audit are real and composed, request bodies are size-bounded (P0-8); OIDC, distributed tracing, and a security incident response procedure are still not built (see §2, Security row).
DEPLOYMENT	GREEN/YELLOW -- deployment guide exists, and since P0-A the Commercial API's Store is durable (WAL-backed, crash-recovering, with a proven backup/restore round trip). Remaining: no operational backup drill against a live deployment, no SLA/SLO draft (see §2, Deployment row).
REAL eBL	EXTERNAL PILOT
REAL AIS	EXTERNAL PILOT
REAL INSURANCE	EXTERNAL PILOT
PEN TEST	PENDING
MULTI-REGION	PENDING
72H SOAK	PENDING
INDEPENDENT AUDIT	PENDING
MLETR LEGAL QUALIFICATION	PENDING
OVERALL	See <code>docs/VERIQO_READINESS_TIER_FRAMEWORK.md</code> for the precise, per-tier verdict. In short: DEMO-READY yes; DESIGN-PARTNER READY yes; PAID-PILOT READY conditionally (all five gating items closed, named operational/documentation conditions remain); NOT PRODUCTION QUALIFIED . Revision note: this row previously read "PILOT-READY, NOT YET FULLY PRODUCTION-QUALIFIED" -- a single phrase that collapsed four commercially distinct commitment levels, correctly criticized as imprecise in review. It is replaced, not softened.

2. Commercial Readiness Matrix (Item 13)

Capability	Status	Evidence
Core Trust Pipeline (Decision Trust Boundary, Trust Calculus, DARI guarantees, deterministic execution, ledger primitives)	GREEN	Frozen per <code>docs/VERIQO_CORE_TRUST_KERNEL_FREEZE.md</code> ; unchanged this sprint except two minimal, behavior-preserving exported-helper extractions (<code>manifest.VerifyCustodyChainRecords</code> , <code>claimworkflow.FinalizeManifestSpec</code>), each re-verified against the full pre-existing test suite.
Evidence Fabric (canonical <code>EvidenceRecord</code> , 3 evidence classes)	GREEN	<code>pkg/commercial/evidencefabric -- FromManifest/FromRegistry</code> , 8 tests including tamper detection (<code>TestFromManifestDetectsTamperedManifest</code>) and all-3-domain-classes coverage.
Decision Trust (grounded Finding -> Decision)	GREEN	Reused verbatim from the frozen <code>cre/decision</code> packages; <code>commercialapi.Store.DecideCase</code> proven to refuse ungrounded evidence (<code>TestDecideCaseRejectsUngroundedEvidence</code>).
Action Authorization	GREEN	Reused verbatim from the frozen <code>action</code> package; <code>ActOnCase</code> proven to refuse acting before a decision exists (<code>TestActOnCaseRefusesBeforeDecision</code>).
Dossier (Evidence Dossier v1)	GREEN	<code>pkg/commercial/dossier -- every reviewer-named field, human Markdown + machine .zip</code> , real Corroboration/Contradictions derived from the actual grounded Finding (not left empty -- see <code>TestGenerateDossierPopulatesRealCorroborationAndContradictions</code>).
Replay	GREEN	<code>verticalslice.Replay</code> and <code>Store.Replay</code> , both proven to converge on identical hashes for unchanged inputs and diverge for a mutated one.
Independent Verifier	GREEN	<code>pkg/commercial/packageverify + cmd/veriqo-commercial-verify --</code> a genuinely separate compiled binary, tested via <code>exec.Command</code> against real and tampered packages, honestly <code>SKIP</code> s signature verification (no signing scheme exists in this reference build) rather than silently passing it.
Commercial API v1	GREEN	All 11 named routes + <code>GET /v1/metrics</code> , wired into <code>NewServer</code> , proven end-to-end over real <code>net/http</code> (<code>TestCommercialV1RoutesFullLifecycleHappyPath</code> : submit evidence -> verify -> custody -> decide -> act -> dossier (JSON + package) -> independently verify the package -> replay).
Tenant Isolation	GREEN	Structural, key-namespaced isolation inside <code>Store</code> is real (<code>TestTenantAIsolationFromTenantB</code> , <code>TestCommercialV1RoutesTenantIsolationOverHTTP</code>). The prior round's named gap is CLOSED (P0-B) : <code>effectiveTenantID</code> derives the acting tenant from the verified JWT subject via <code>pkg/commercial/tenancy.Membership</code> , refusing (403) any tenant that subject is not granted -- <code>commercial_v1_tenant_binding_test.go</code> (5 tests, real signed JWTs), <code>pkg/commercial/tenancy/tenancy_test.go</code> (6 tests).
Security (JWT/RBAC/API keys/audit)	GREEN/YELLOW	Real and composed (<code>security.JWTMiddleware/RBACMiddleware/APIKeyMiddleware/AuditMiddleware</code>), plus request-body size bounds on every Commercial API v1 POST route (P0-8, <code>http.MaxBytesReader</code> , <code>TestOversizedRequestBodyIsRejected</code>). Remaining gaps : no OIDC, no distributed tracing, no written incident response procedure.
Observability (metrics/logs/audit/health)	GREEN/YELLOW	<code>pkg/commercial/telemetry</code> 's 9 named counters are real (<code>TestMetricsReflectRealActivity</code>); logging and audit are real. Liveness/readiness probes CLOSED (P0-6) : <code>GET /livez</code> (dependency-free) and <code>GET /readyz</code> (reports 503 on a closed durable Store, proven by <code>TestReadyzReportsNotReadyOnceCommercialStoreIsClosed</code>). Remaining gaps : no traces, no alerting.
Deployment	GREEN/YELLOW	<code>docs/governance/PRODUCTION_DEPLOYMENT_AND_OPERATIONS_GUIDE.md</code> exists for the broader platform. The prior round's named gap is CLOSED (P0-A) : <code>NewDurableStore</code> makes <code>commercialapi.Store</code> durable over the real <code>pkg/storage/wal</code> (fsync, CRC, defect-classified recovery), with <code>Backup/RestoreStoreFromBackup</code> and a proven round trip -- <code>TestNewDurableStoreReconstructsIdenticalStateAfterRestart</code> , <code>TestDurableStoreRecoversFromATornWrite</code> , <code>TestStoreBackupAndRestoreRoundTrip</code> , plus a live restart test against the compiled gateway binary. Remaining : no operational backup drill against a live deployment, no SLA/SLO draft.
Demo Cases (eBL, Maritime, Insurance)	GREEN	<code>pkg/commercial/democases + cmd/veriqo-demo-cases</code> ; all 3 build real, independently-verifiable packages -- see <code>docs/VERIQO_DEMO_CASES.md</code> .
Real eBL / Real AIS / Real Insurance integrations	EXTERNAL PILOT	Per item 11, real insurer/adjuster/P&I/AIS/eBL-platform integrations are explicitly out of this engagement's scope -- they require external counterparties and data-sharing agreements this engagement cannot create.

Capability	Status	Evidence
Pen Test / Multi-Region / 72h Soak / Independent Audit	PENDING	Mechanism proven (<code>docs/governance/production-blockers.json</code> , all 8 blockers <code>READY_FOR_REAL_QUALIFICATION</code>); real external qualification (an actual vendor engagement, actual multi-region infra, actual 72-hour production-scale run) has not happened.
MLETR Legal Qualification	PENDING	<code>docs/VERIQO_MLETR_EBL_CONFORMANCE_MAPPING_V0_2.md</code> records an honest, repeated primary-source-verification attempt that could not reach a legal authority able to confirm MLETR conformance from this engagement's position -- this remains a qualified legal opinion this engagement cannot produce.

3. Documentation Inventory (Item 21)

Rather than fabricate placeholder documents for every item named, this inventory reports **what already exists and covers the item**, what is **partially covered**, and what is a **genuine gap** -- consistent with this engagement's discipline of never claiming a deliverable exists when it does not.

Technical documentation

Named document	Status	Where
Architecture Overview	DONE	<code>docs/ARCHITECTURE.md</code>
API Specification	PARTIAL	Every Commercial API v1 route is documented in <code>pkg/commercial/api/store.go</code> 's and <code>commercial_v1_routes.go</code> 's own doc comments (request/response shapes, status-code mapping) -- no single standalone OpenAPI/API-spec document exists yet.
Security Model	DONE	<code>docs/THREAT_MODEL.md</code> , <code>docs/governance/SECURITY_QUALIFICATION_PACK.md</code>
Evidence Model	PARTIAL	<code>pkg/commercial/evidencefabric</code> 's own doc comment is the canonical model description; no standalone customer-facing "Evidence Model" document exists.
Trust Model	DONE	<code>docs/VERIQO_TRUST_AUTHORITY_MODEL_RESPONSE.md</code>
Replay Specification	PARTIAL	<code>verticalslice.Replay/Store.Replay</code> 's own doc comments describe the mechanism; no standalone spec document.
Verifier Specification	PARTIAL	<code>docs/governance/CASE_ROOM_AND_DOSSIER_VERIFIER_SPECIFICATION.md</code> covers the earlier insurance-domain verifier; <code>pkg/commercial/packageverify</code> 's own doc comment covers this sprint's Commercial verifier, but the two have not been consolidated into one spec.
Deployment Guide	DONE	<code>docs/governance/PRODUCTION_DEPLOYMENT_AND_OPERATIONS_GUIDE.md</code>
Backup/Restore	PARTIAL	Mechanism now exists (P0-A): <code>Store.Backup/RestoreStoreFromBackup</code> with a proven round trip (<code>TestStoreBackupAndRestoreRoundTrip</code>) -- backup and normal crash recovery share one code path by design, so a backup is not a separate untested format. Remaining gap: no written operator runbook, and no drill against a live deployment.
Incident Response	GAP	No document exists. Named as a MUST-CLOSE-BEFORE-PAID-PILOT item in <code>docs/VERIQO_PILOT_MODE_AND_DEPLOYMENT_READINESS.md</code> .

Customer-facing documentation

Named document	Status	Where
VERIQO Overview	GAP	No customer-facing (non-technical) overview document exists; <code>docs/ARCHITECTURE.md</code> is written for engineers.
Pilot Guide	GAP	No document exists.
Case Walkthrough	DONE	<code>docs/VERIQO_DEMO_CASES.md</code> -- three full walkthroughs in the exact What-Knows/Does-Not-Know/Can-Prove/Legal-Question shape item 14-16 ask for.
Evidence Dossier Guide	PARTIAL	<code>dossier.RenderMarkdown</code> 's field-by-field output and <code>docs/VERIQO_DEMO_CASES.md</code> 's worked examples show what a dossier contains; no standalone "how to read your dossier" guide.

Named document	Status	Where
Integration Guide	GAP	No customer-facing "how to call the Commercial API v1 from your systems" guide exists (distinct from the internal API Specification gap above).
Security FAQ	GAP	No document exists.
Data Handling	PARTIAL	<code>docs/data/VERIQO_EXTERNAL_DATA_RIGHTS_REGISTER.md</code> covers data-rights obligations for external sources; no customer-facing data-handling document.
Limitations	PARTIAL	Every dossier's <code>standardLimitations</code> (see <code>pkg/commercial/dossier/dossier.go</code>) and <code>docs/VERIQO_DEMO_CASES.md</code> 's "Remains a Legal Question" sections state real limitations per-case; no single consolidated Limitations document.
Legal positioning	GAP	Item 8's marketing-language discipline ("Court/Arbitration Evidence Support," never "Court-admissible guaranteed") is followed in every document this engagement writes, but no standalone legal-positioning document exists.
MLETR positioning	DONE	<code>docs/VERIQO_MLETR_EBL_CONFORMANCE_MAPPING_V0_2.md</code>
Electronic evidence positioning	PARTIAL	Covered inside the MLETR mapping and <code>docs/THREAT_MODEL.md</code> ; no standalone document.
No-legal-conclusion policy	PARTIAL	Enforced in every dossier's Limitations section and this engagement's own language discipline; no standalone policy document.
Jurisdiction disclaimer	GAP	No document exists.

Why these gaps are named, not filled: several of the customer-facing gaps above (VERIQO Overview, Pilot Guide, Security FAQ, Legal positioning, Jurisdiction disclaimer) are sales/legal collateral that item 26 explicitly places AFTER the readiness gate, as part of business development this engagement is not asked to execute. Writing hollow placeholder versions of them now would be exactly the kind of unsubstantiated claim this engagement's discipline refuses to make; they are named here as real, specific next steps instead.

4. Commercialization Gate (Item 22)

Gate	Scope	Status
GATE A -- Core	Frozen Core Trust Kernel (Decision Trust Boundary, ActionAuthorization, Trust Calculus, DARI, ledger primitives)	PASS -- frozen, unchanged except minimal API stabilization, fully re-verified.
GATE B -- Evidence Fabric	Canonical <code>EvidenceRecord</code> , vertical slice <code>SOURCE->RECEIPT</code>	PASS -- <code>pkg/commercial/evidencefabric</code> + <code>pkg/commercial/verticalslice</code> , both tested.
GATE C -- Dossier & Verification	Evidence Dossier v1, independent standalone verifier	PASS -- <code>pkg/commercial/dossier</code> , <code>pkg/commercial/packageverify</code> , <code>cmd/veriqo-commercial-verify</code> .
GATE D -- Commercial API	The 11 named <code>/v1/...</code> routes, no internal kernel types exposed	PASS -- <code>pkg/commercial/api</code> + <code>veriqo/gateway/rest/commercial_v1_routes.go</code> , full HTTP-level lifecycle test.
GATE E -- Pilot Mode & Security	Tenant isolation, RBAC/JWT/API-key, audit, observability	PASS WITH NAMED GAPS -- see §2's YELLOW rows; nothing here is silently assumed complete.
GATE F -- Demonstrability	3 canonical demo cases, customer-facing case walkthrough	PASS -- <code>pkg/commercial/democases</code> , <code>docs/VERIQO_DEMO_CASES.md</code> .
GATE G -- Production	HSM/KMS, durable ledger, production identity/mTLS, secure object storage, vulnerability scanning, backup/restore, production monitoring, incident response, multi-region, 100-node scale, 72h soak, independent pen test	NOT YET -- every mechanism this engagement can build without external infrastructure is built and proven (<code>READY_FOR_REAL_QUALIFICATION</code> across all 8 tracked blockers); real external qualification has not happened. Per item 10's own instruction, this is reported as <code>NOT_YET_PRODUCTION_QUALIFIED</code> , not hidden.

5. What's Next (Item 26 -- Not This Engagement's Job)

Per the reviewer's own item 26, the following belong to business development once a real pilot customer is engaged, and are explicitly outside what this engagement executes: customer acquisition, pilot proposal, data-sharing agreements, NDAs, paid-pilot commercial structuring, pricing, security questionnaires, customer onboarding, ROI measurement against real pilot data, and external data contracts (for the real eBL/AIS/insurance integrations named EXTERNAL PILOT above). This report's job ends at handing over an honest gate assessment; it does not attempt any of the above.

6. Bottom Line

The single-phrase verdict this section previously carried ("PILOT-READY, NOT YET FULLY PRODUCTION-QUALIFIED") has been replaced by an explicit four-tier assessment -- see docs/VERIQO_READINESS_TIER_FRAMEWORK.md, which is now the single source of truth for VERIQO's readiness position:

- **DEMO-READY: YES.**
- **DESIGN-PARTNER READY: YES.**
- **PAID-PILOT READY: CONDITIONALLY.** All five gating items a first paying customer would refuse to pay without are now closed -- durable persistence (P0-A), tenant identity binding (P0-B), evidence retention/legal hold (P0-C), cryptographic signing (P0-D), and an independent verifier that performs real signature/key-state verification rather than a permanent SKIP (P0-E). The remaining conditions are operational and documentation work, enumerated exhaustively in the tier framework's "Conditions Still Open" section (HTTP surfaces for the retention/signing capabilities, OIDC, incident response procedure, SLA/SLO draft, an operational backup drill, HSM-grade key custody, and an external security review).
- **PRODUCTION QUALIFIED: NO.** All 8 tracked blockers read `READY_FOR_REAL_QUALIFICATION`; the real external qualification (pen test, HSM/KMS tenancy, multi-region, 72-hour production-scale soak, independent audit) has not happened, and `READY_FOR_REAL_QUALIFICATION` is explicitly not `QUALIFIED`. MLETR legal qualification likewise remains a legal opinion this engagement cannot produce.

None of the remaining work requires redesigning anything this or any prior sprint built.